Sixdof Space
ULTRA FAST OPTICAL TRACKING

Software Package Options

The Sixdof positioning and tracking systems are sold in two variations. Both the Sixdof Falcon and the Sixdof Cheetah use the Sixdof sensor and beacon hardware to provide high-speed positioning with full 6DOF tracking.

SIXDOF FALCON

The Sixdof Falcon tracking system provides full 6DOF tracking by deploying three or more IR beacons at set locations. The Sixdof sensor (in motion) detects the beacons and calculates its relative position and orientation 400 times a second.

Falcon is the perfect fit for an autonomous system. For example, for landing a drone, the IR beacons can be embedded in the landing pad, and the sensor mounted on the drone facing down. A small onboard companion computer is required to interface between the Sixdof sensor and the drone controller.

With this configuration, the Falcon system can enable full autonomous landing in challenging environments, such as windy conditions or on a moving target.

The Falcon Development Kit Includes:

- One Kipa Sensor Unit
- Shale beacon driver board with four low-powered beacons (7m) and four high-powered beacons (20m)
- Sixdof Falcon SDK
- Up to 10 Hours Remote Support

SIXDOF CHEETAH

The Sixdof Cheetah system leverages one or more sensors at fixed locations to track IR beacons attached to moving targets. With two sensors with a large baseline between them, Cheetah can provide highly accurate tracking data for beacons at a distance.

Additionally, when three or more beacons are affixed to a tracking target, Cheetah can track the target's full 6DOF position and orientation. It's the perfect fit where the tracking data is used on the ground unit, such as a drone capture mechanism.

Cheetah is also the solution in micro-drone applications, where the footprint and power consumption of the sensor and companion computer are too cumbersome. In these cases, a micro-drone can be equipped with the beacons powered by a small battery, resulting in a much smaller payload.

The Cheetah Development Kit Includes:

- Two Kipa Sensor Units
- Shale beacon driver board with six low-power beacons (7m)
- Sixdof Cheetah SDK
- Up to 10 Hours Remote Support

Elevate your robotics and automation capabilities with Sixdof's unmatched optical tracking precision—accelerate your operations today.

For more details, visit us at www.sixdofspace.com